Example 12.6

In this example, we will analyze the small-signal stability of a simple two-area system shown in Figure E12.8. This system is similar in structure to the one used in references 23 and 24 to study the fundamental nature of interarea oscillations.

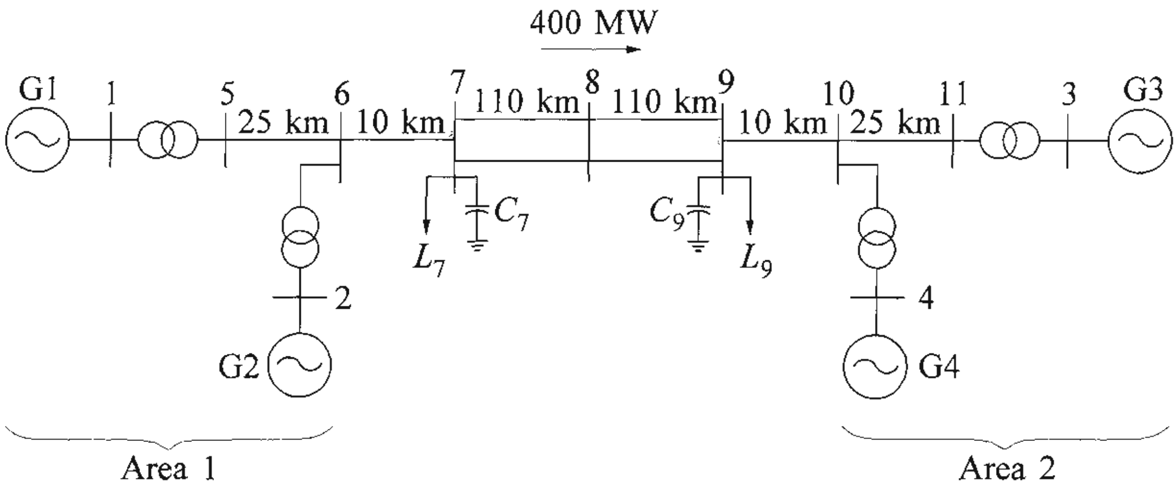


Figure E12.8 A simple two-area system

The system consists of two similar areas connected by a weak tie. Each area consists of two coupled units, each having a rating of 900 MVA and 20 kV. The generator parameters in per unit on the rated MVA and kV base are as follows:

$X_d = 1.8$ $X_q = 1.7$ $X_l = 0.2$ $X_d' = 0.3$ $X_q' = 0.55$
$X_d'' = 0.25$ $X_q'' = 0.25$ $R_a = 0.0025$ $T_{d0}' = 8.0$ s $T_{q0}' = 0.4$ s
$T_{d0}'' = 0.03$ s $T_{q0}'' = 0.05$ s $A_{Sat} = 0.015$ $B_{Sat} = 9.6$ $\psi_{T1} = 0.9$
$H = 6.5$ (for G1 and G2) $H = 6.175$ (for G3 and G4) $K_D = 0$

Each step-up transformer has an impedance of $0+j0.15$ per unit on 900 MVA and 20/230 kV base, and has an off-nominal ratio of 1.0.

The transmission system nominal voltage is 230 kV. The line lengths are identified in Figure E12.8. The parameters of the lines in per unit on 100 MVA, 230 kV base are

$$r = 0.0001 \text{ pu/km} \qquad x_L = 0.001 \text{ pu/km} \qquad b_C = 0.00175 \text{ pu/km}$$

The system is operating with area 1 exporting 400 MW to area 2, and the generating units are loaded as follows:

G1:	$P = 700$ MW,	$Q = 185$ MVAr,	$E_t = 1.03\angle 20.2°$
G2:	$P = 700$ MW,	$Q = 235$ MVAr,	$E_t = 1.01\angle 10.5°$
G3:	$P = 719$ MW,	$Q = 176$ MVAr,	$E_t = 1.03\angle -6.8°$
G4:	$P = 700$ MW,	$Q = 202$ MVAr,	$E_t = 1.01\angle -17.0°$

The loads and reactive power supplied (Q_C) by the shunt capacitors at buses 7 and 9 are as follows:

Bus 7: P_L = 967 MW, Q_L = 100 MVAr, Q_C = 200 MVAr
Bus 9: P_L = 1,767 MW, Q_L = 100 MVAr, Q_C = 350 MVAr

(a) If all four generators are on manual excitation control (constant E_{fd}), compute the eigenvalues of the system state matrix representing the small-signal performance of the system about the initial operating condition. For each eigenvalue, identify the system state variables with high participation. Determine the frequencies, damping ratios, and mode shapes of the rotor oscillation modes. Assume that the active components of loads have constant current characteristics, and reactive components of loads have constant impedance characteristics.

(b) Determine the eigenvalues, frequencies, and damping ratios of rotor oscillation modes when all four generators are equipped with the following types of excitation control:

(i) Self-excited dc exciter (see Figures 8.38 and 8.40):

$K_A = 20.0$ $T_A = 0.055$ $T_E = 0.36$ $K_F = 0.125$
$T_F = 1.8$ $A_{ex} = 0.0056$ $B_{ex} = 1.075$ $T_R = 0.05$
T_B, T_C, R_C, and X_C are not used

(ii) Thyristor exciter with a high transient gain:

$K_A = 200.0$ $T_R = 0.01$

(iii) Thyristor exciter with a transient gain reduction (TGR):

$K_A = 200.0$ $T_R = 0.01$ $T_A = 1.0$ $T_B = 10.0$

(iv) Thyristor exciter with high transient gain and PSS:

$K_A = 200.0$ $T_R = 0.01$ $K_{STAB} = 20.0$ $T_W = 10.0$
$T_1 = 0.05$ $T_2 = 0.02$ $T_3 = 3.0$ $T_4 = 5.4$

The block diagram of thyristor excitation system with PSS is shown in Figure E12.9.

Solution

(a) *System modes with generators on manual excitation control:*

Table E12.3 summarizes the eigenvalues of the system state matrix and the state variables that have high participation in each mode. The first two eigenvalues represent the *zero eigenvalues* due to the redundant state variables. As described in Section 12.7, one of these zero eigenvalues is due to lack of uniqueness of absolute rotor angle (there is no infinite bus, and the rotor angles are referred to a common reference frame). The other zero eigenvalue is due to the assumption that the generator torques are independent of speed deviation (speed governors are not modelled and K_D=0).

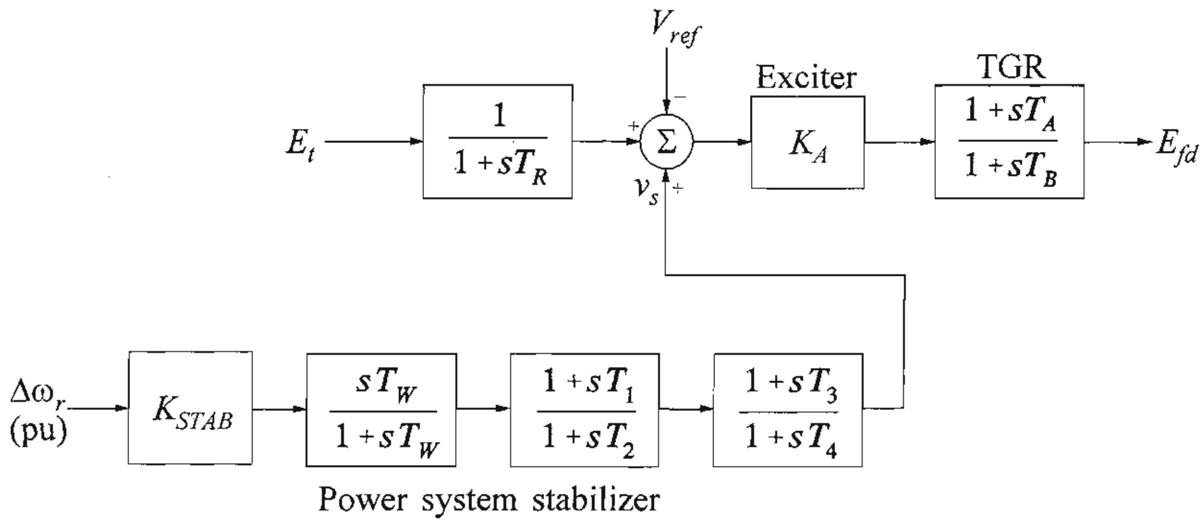
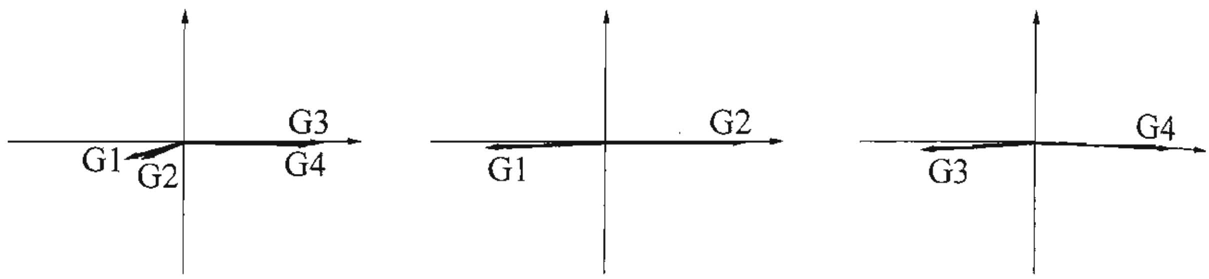


Figure E12.9 Thyristor excitation system with PSS

Table E12.3 System modes with manual excitation control

Eigenvalues			Frequency (Hz)	Damping Ratio	Dominant States
No.	Real	Imaginary			
1,2	-0.76E-3	±0.22E-2	0.0003	0.331	$\Delta\omega$ and $\Delta\delta$ of G1, G2, G3, G4
3	-0.96E-1	–	–	–	"
4,5	-0.111	±3.43	0.545	0.032	"
6	-0.117	–	–	–	"
7	-0.265	–	–	–	$\Delta\psi_{fd}$ of G3 and G4
8	-0.276	–	–	–	$\Delta\psi_{fd}$ of G1 and G2
9,10	-0.492	±6.82	1.087	0.072	$\Delta\omega$ and $\Delta\delta$ of G1 and G2
11,12	-0.506	±7.02	1.117	0.072	$\Delta\omega$ and $\Delta\delta$ of G3 and G4
13	-3.428	–	–	–	
14	-4.139	–	–	–	*d* and *q* axis amortisseur flux linkages
15	-5.287	–	–	–	
16	-5.303	–	–	–	
17	-31.03	–	–	–	
18	-32.45	–	–	–	
19	-34.07	–	–	–	
20	-35.53	–	–	–	
21,22	-37.89	±0.142	0.023	≈1.0	
23,24	-38.01	±0.38E-1	0.006	≈1.0	

From the table, we see that the system is stable. There are three rotor angle modes of oscillation. Their mode shapes (normalized eigenvector components corresponding to rotor speeds of the four machines) are shown in Figure E12.10. From the mode shapes we see that the 0.55 Hz mode is the interarea mode, with generators G1 and G2 of area 1 swinging against generators G3 and G4 of area 2. The 1.09 Hz mode is the intermachine oscillation local to area 1, with G1 swinging against G2. The third rotor angle mode, with a frequency of 1.12 Hz, is the intermachine mode local to area 2.



(a) Interarea mode f=0.545 Hz, ζ=0.032 (b) Area 1 local mode f=1.087 Hz, ζ=0.072 (c) Area 2 local mode f=1.117 Hz, ζ=0.072

Figure E12.10 Mode shapes of rotor angle modes with manual excitation control

(b) *Rotor angle modes with different types of excitation control:*

Table E12.4 summarizes the eigenvalues, frequencies, and damping ratios associated with the rotor oscillation modes for the four alternative forms of excitation control.

We see that the local intermachine modes of oscillation have the same degree of damping with the dc exciter and thyristor exciter (with and without TGR). The interarea mode has a small positive damping with the dc exciter. It is unstable with high gain thyristor exciters. The TGR makes the interarea mode more unstable. The interarea mode as well as the local modes are very well damped when power system stabilizers are added to the thyristor exciters.

Table E12.4 Effects of excitation control on rotor oscillation modes

Type of Excitation Control	Eigenvalue/(Frequency in Hz, Damping Ratio)		
	Interarea Mode	Area 1 Local Mode	Area 2 Local Mode
(i) DC exciter	$-0.018 \pm j3.27$ (f=0.52, ζ=0.005)	$-0.485 \pm j6.81$ (f=1.08, ζ=0.07)	$-0.500 \pm j7.00$ (f=1.11, ζ=0.07)
(ii) Thyristor with high gain	$+0.031 \pm j3.84$ (f=0.61, ζ=−0.008)	$-0.490 \pm j7.15$ (f=1.14, ζ=0.07)	$-0.496 \pm j7.35$ (f=1.17, ζ=0.07)
(iii) Thyristor with TGR	$+0.123 \pm j3.46$ (f=0.55, ζ=−0.036)	$-0.450 \pm j6.86$ (f=1.09, ζ=0.06)	$-0.462 \pm j7.05$ (f=1.12, ζ=0.06)
(iv) Thyristor with PSS	$-0.501 \pm j3.77$ (f=0.60, ζ=0.13)	$-1.826 \pm j8.05$ (f=1.28, ζ=0.22)	$-1.895 \pm j8.35$ (f=1.33, ζ=0.22)

■